

Syllabus

Ad-hoc and sensor networks

Module 1: Introduction

Fundamentals of wireless communication technology the electromagnetic spectrum radio propagation mechanisms characteristics of the wireless channel – Mobile Ad-hoc Networks (MANETS) and Wireless Sensor Networks (WSNs): concepts and architectures. Applications of Ad-hoc and sensor networks. Design challenges in Ad-hoc and sensor networks.

Module 2:

Mac Protocols for Ad-hoc Wireless Networks Issues in designing a MAC Protocol- Classification of MAC Protocols-Contention based protocols Contention based protocols with Reservation Mechanisms- Contention based protocols with Scheduling Mechanisms – Multi channel MAC-IEEE 802.11

Module 3:

Routing Protocols and Transport Layer in Ad-hoc Networks. Issues in designing a routing and Transport Layer protocol for Ad hoc networks- proactive routing, reactive routing (on-demand), hybrid routing- Classification of Transport Layer solutions-TCP over Ad hoc wireless Networks.

Module 4:

Wireless Sensor Networks (WSNs) And MAC Protocols. Single node architecture: hardware and software components of a Sensor node WSN Network architecture: typical network architectures-data relaying and aggregation strategies -MAC layer protocols: self-organizing, Hybrid TDMA/FDMA and CSMA based MAC-IEEE 802.15.4

Module 5:

Security in Ad Hoc and Sensor Networks Security Attacks Key Distribution and Management Intrusion Detection – Software based Anti-tamper techniques – Water marking techniques – Defence against routing attacks – Secure Ad hoc routing protocols – Broadcast authentication WSN protocols – TESLA – Biba Sensor Network Security Protocols – SPINS

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Digital Image Processing

Module 1: Introduction:

Background, Digital Image Representation, Fundamental Steps in Image Processing, Elements of a Digital Image Processing System. Digital Image Fundamentals: Elements of Visual Perception, a Simple Image Model, Sampling and Quantization, Some Basic Relationships between Pixels, Imaging Geometry.

Module 2

Image Transforms: Introduction to the Fourier Transform, The Discrete Fourier Transform, Some Properties of the Two-Dimensional Fourier Transform, Other Separable Image Transforms.

Module 3:

Image Enhancement: Spatial Domain Methods, Frequency Domain Methods, Some Simple Intensity Transformations, Histogram Processing, Image Subtraction, Image Averaging, Background, Smoothing Filters, Sharpening Filters, Lowpass Filtering, Highpass Filtering, Generation of Spatial Masks from Frequency Domain Specifications.

Module 4

Image Restoring: Degradation Model Definitions, Degradation Model for Continuous Functions, Diagonalization of Circulant and Block-Circulant Matrices, Circulant Matrices, Block Circulant Matrices, Effects of Diagonalization on the Degradation Model, Algebraic Approach to Restoration, Unconstrained Restoration, Constrained Restoration, Inverse Filtering – Formulation, Removal of Blur Caused by Uniform Linear Motion, Restoration in the Spatial Domain, Geometric Transformation.

Module 5

Image Compression: Fundamentals – Coding Redundancy, Interpixel Redundancy, Psychovisual Redundancy, Fidelity Criteria. Image Compression Models The Source Encoder and Decoder, The Channel Encoder and Decoder. Elements of Information Theory Measuring Information, The Information Channel, Fundamental Coding Theorems, Using Information Theory. Error-Free Compression Variable-Length Coding, Bit-Plane Coding, Lossless Predictive Coding. Lossy Compression Lossy Predictive Coding, Transform Coding.

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DATA MINING

Module I

Data Warehousing and Business Analysis: - Data warehousing Components-Building a Data warehouse-Data Warehouse Architecture DBMS Schemas for Decision Support - Data Extraction, Cleanup, and Transformation Tools -Metadata reporting Query tools and Applications Online Analytical Processing (OLAP) - OLAP and Multidimensional Data Analysis.

Module 2

Data Mining: Data Mining Functionalities Data Preprocessing - Data Cleaning Data Integration and Transformation - Data Reduction Data Discretization and Concept Hierarchy Generation-Architecture Of A Typical Data Mining Systems-Classification Of Data Mining Systems. Association Rule Mining: Efficient and Scalable Frequent Item set Mining Methods Mining Various Kinds of Association Rules - Association Mining to Correlation Analysis Constraint-Based Association Mining.

Module 3

Classification and Prediction: - Issues Regarding Classification and Prediction Classification by Decision Tree Introduction Bayesian Classification Rule Based Classification Classification by Back propagation Support Vector Machines Associative Classification - Lazy Learners Other Classification Methods Prediction - Accuracy and Error Measures - Evaluating the Accuracy of a Classifier or Predictor Ensemble Methods Model Section.

Module 4

Cluster Analysis: Types of Data in Cluster Analysis - A Categorization of Major Clustering Methods Partitioning Methods - Hierarchical methods Density-Based Methods Grid-Based Methods Model-Based Clustering Methods Clustering High-Dimensional Data Constraint-Based Cluster Analysis - Outlier Analysis.

Module 5 .

Mining Object, Spatial, Multimedia, Text and Web Data: Multidimensional Analysis and Descriptive Mining of Complex Data Objects Spatial Data Mining Multimedia Data Mining - Text Mining Mining the World Wide Web.

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BLOCK CHAIN

Module I

Introduction: Overview of Block chain, Public Ledgers, Bitcoin, Smart Contracts, Block in a Block chain, Transactions, Distributed Consensus, Public vs Private Block chain, Understanding Crypto currency to Block chain, Permissioned Model of Block chain, Overview of Security aspects of Block chain Basic Crypto Primitives: Cryptographic Hash Function, Properties of a hash function, Hash pointer and Merkle tree, Digital Signature, Public Key Cryptography, A basic cryptocurrency.

Module 2

Understanding Block chain with Crypto currency: Bitcoin and Block chain: Creation of coins, Payments and double spending. Bitcoin Scripts, Bitcoin P2P Network, Transaction in Bitcoin Network, Block Mining, Block propagation and block relay.

Working with Consensus in Bitcoin: Distributed consensus in open environments, Consensus in a Bitcoin network, Proof of Work (POW) - basic introduction, Hashcash Pow, Bitcoin Pow, Attacks on Pow and the monopoly problem, Proof of Stake, Proof of Burn and Proof of Elapsed Time, The life of a Bitcoin Miner, Mining Difficulty, Mining Pool.

Module 3

Understanding Block chain for Enterprises: Permissioned Block chain: Permissioned model and use cases, Design issues for Permissioned block chains, Execute contracts, State Machine replication, Overview of Consensus Models for permissioned block chain- Distributed consensus in closed environment, Paxos, RAFT Consensus, Byzantine general problem, Byzantine fault tolerant system, Lamport-Shostak-Pease BFT Algorithm, BFT over Asynchronous systems.

Enterprise application of Block chain: Cross border payments, Know Your Customer (KYC), Food Security, Mortgage over Block chain, Block chain enabled Trade, We Trade Trade Finance Network, Supply Chain Financing, Identity on Block chain

Module 4 .

Block chain application development: Hyperledger Fabric-Architecture, Identities and Policies, Membership and Access Control, Channels, Transaction Validation, Writing

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smart contract using Hyperledger Fabric, Writing smart contract using Ethereum,
Overview of Ripple and Corda.